

Reading the NOR140

Every measurement this system records, what it physically means, and how it is derived from the raw bytes on the meter's SD card. Written for someone doing this work for the first time.

Norsonic NOR140 · Class 1 to IEC 61672-1:2013 · 1/3-octave, 6.3 Hz – 20 kHz · configured here for one profile sample per second

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What a single run actually contains

A "run" is one continuous measurement, typically 15 minutes. A "session" is all the runs from one date.

The meter writes two binary files per run. `GLOBnnnn.DAT` holds the summary of the whole run — single numbers and frequency spectra. `PROFnnnn.DAT` holds a time profile: five values recorded every second for the entire run.

22

broadband results — 11
quantities × 2 weightings

16

percentiles — 8×2
weightings

648

spectral values — 18
tables × 36 bands

4,500

one-second samples — 5
channels × 900 s

That is roughly **5,200 numbers per 15-minute run**. They are not 5,200 independent facts — most are the same sound pressure viewed through different filters. Learning which filter each one applies is the whole job, and it is much smaller than the count suggests.

Notice the doubling in the first two figures. The 22 broadband results and the 16 percentiles are stored as **A/C pairs** — the same quantity filtered two different ways. The other two groups are not doubled: the 648 spectral values are a single *unweighted* set, and the 4,500 profile samples are A-weighted only. So the pairing covers 38 of the roughly 5,200 values, but those 38 are the ones you report, which is why A and C run through everything that follows. They are the second of the three ideas below.

Three ideas first

Almost every metric is a combination of these three choices. Get them straight and the catalogue reads itself.

1. Decibels are logarithmic, so you cannot average them normally

A decibel is a ratio on a log scale: +10 dB is ten times the sound energy, +3 dB is roughly double. This has a consequence people get wrong constantly — you cannot take the arithmetic mean of two dB values and get a meaningful answer. To combine levels you must convert back to energy, average that, and convert back:

$$Leq = 10 \cdot \log_{10} \left(\frac{1}{n} \cdot \sum 10^{(L_i/10)} \right) \text{ energy average — correct mean = } \\ (L_1 + L_2 + \dots + L_n) / n \text{ arithmetic mean — wrong for dB}$$

Averaged over equal durations, 60 dB and 70 dB together are 67.4 dB, not 65 dB. The louder sound dominates, because it carries ten times the energy. When the periods are *not* equal, each term has to be weighted by its duration — which is exactly what this application does when combining runs into a session.

2. Frequency weighting — which pitches you count

Human hearing is far less sensitive to low frequencies than to speech frequencies. A weighting curve applies that bias before anything is measured, so the number reflects what a person perceives rather than raw physics.

FREQUENCY WEIGHTINGS					
CODE	NAME	63 HZ	1 KHZ	8 KHZ	USED FOR
A	A-weighted	-26 dB	0 dB	-1 dB	Almost everything. Matches perceived loudness.
C	C-weighted	-1 dB	0 dB	-3 dB	Peaks and impulses, where low-frequency energy matters.
Z	Zero / unweighted	0 dB	0 dB	0 dB	Raw physical level. The stored 1/3-octave spectra are unweighted like this; the stored broadband scalars are all A or C.

Why two curves, and not just one? They answer different questions. **A-weighting** approximates hearing sensitivity, and carries most environmental and occupational average-level criteria — it deliberately discards low-frequency energy the ear barely registers. **C-weighting** is far flatter and retains much more low-frequency content, which makes it the right form for peak criteria and for diagnosing low-frequency sources.

Neither is raw physical pressure. C is a weighting too, and still a substantial one at the edges: it discards **21 dB at 6.3 Hz and 11 dB at 20 kHz** — the exact bottom and top of this meter's own range. Z-weighting is the closest thing to unweighted pressure, and it is the three of them, not two, that are genuinely distinct.

You get both because the instrument runs both networks at once: Norsonic specifies simultaneous A and C, or A and Z, measurement. The pair is not computed afterwards from one signal — it is measured in parallel, which is why every broadband result arrives already in two forms.

Two curves are needed because the ear itself changes with level. At ordinary levels it is quite insensitive to low frequencies, which is what A-weighting models. At very high levels its response flattens out, so a loud impulse is better represented by the flatter curve — which is why peak limits are written C-weighted and averaged exposure limits are written A-weighted. Storing both is what lets the pair be more informative than either alone: the *gap* between them is itself the diagnostic for how low-frequency a sound is.

A WEIGHTING IS NOT A FREQUENCY BAND

Both live in the frequency domain, so they are easy to confuse. A **weighting** is one curve applied across the entire spectrum to produce a single number — "A-weighted" describes how a level was filtered, not which pitches it covers. A **band** is a slice of the spectrum; this meter uses 36 one-third-octave bands from 6.3 Hz to 20 kHz, and you meet them in the spectra later. So LAeq is not "the low band" or "the speech band" — it is the whole sound, filtered to approximate hearing.

Because A-weighting throws away roughly 26 dB at 63 Hz where C-weighting keeps almost all of it, most real sounds give a higher C-weighted number than A-weighted. Any metric whose C value sits far above its A value is telling you the sound is low-frequency.

NOT A UNIVERSAL RULE

C is *not* always higher. Above about 1 kHz the curves cross: A-weighting adds up to +1.3 dB around 2.5 kHz while C-weighting is already going negative, so by 5 kHz A sits 1.8 dB *above* C. A sound dominated by mid-to-high frequencies can therefore have LAeq greater than LCeq — 10 of the 527 runs in this project's archive do exactly that. Treat a high C-minus-A difference as evidence of low-frequency content, not as a law.

3. Time weighting — how fast the meter reacts

Sound level is a moving target, so the meter reports a continuously smoothed version of it. Each detector is an *exponentially time-weighted RMS* — a rolling root-mean-square with an exponential decay rather than a flat window — and the time constant of that decay is the time weighting. It changes the answer substantially for anything short and sharp.

TIME WEIGHTINGS			
CODE	NAME	TIME CONSTANT	BEHAVIOUR
F	Fast	125 ms	The default. Follows normal events without chasing every transient.
S	Slow	1 s	Heavily smoothed. Usually gives the lowest of the three maxima for transient sounds; on steady sounds the three converge.
I	Impulse	35 ms rise	Rises fast, decays slowly. Designed to catch bangs and impacts.
peak	Peak (no averaging)	—	The true instantaneous crest of the waveform. Not a time weighting at all.

THE DISTINCTION THAT MATTERS MOST

Peak is not a kind of maximum. A maximum is the highest value of an *averaged* level. A peak is the highest instantaneous pressure, with no averaging whatsoever. Peak is therefore always the larger number, and the gap between them tells you how impulsive the noise was — see [the trap below](#).

Reading a metric name

Every symbol is built from the same four slots. Once you can parse the name, you never need to look up a metric again.

LAFmax

L	A	F	MAX
Level, in decibels	Frequency weighting A · C · Z	Time weighting F · S · I	Statistic over the run max · min · eq · E · 90

LAFmax is therefore "the highest Fast-averaged, A-weighted level reached during the run". Read the same way: **LCpeak** is the C-weighted instantaneous crest, **LAEq** is the A-weighted energy average, **LA90** is the A-weighted level exceeded 90% of the time.

THE STATISTIC SLOT

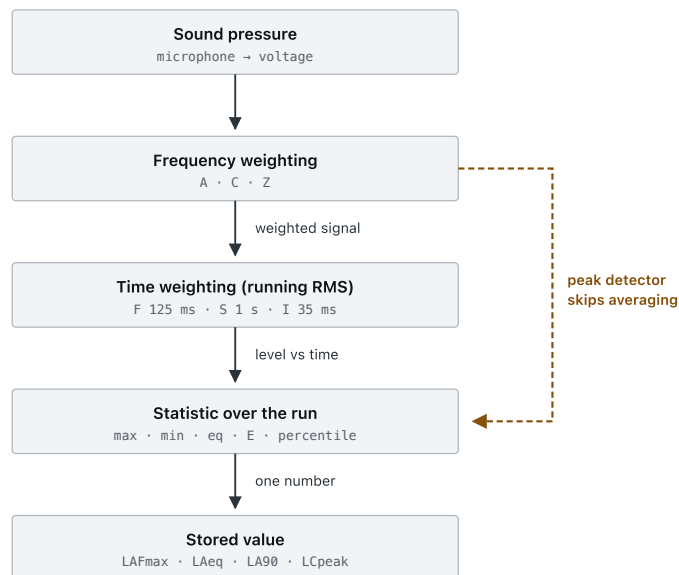
SLOT	MEANING	PLAIN ENGLISH
eq	Equivalent continuous level	The steady level carrying the same total energy as the real, varying sound. The single most important number.
max	Maximum	Loudest moment, as seen through the time weighting.
min	Minimum	Quietest moment. Often a proxy for the true background.
peak	Peak	Highest instantaneous pressure. No averaging.
E	Sound exposure level	All the run's energy compressed into one second. Lets you compare events of different lengths.
10 / 50 / 90	Percentile	The level exceeded that percentage of the time. LA90 indicates background, LA10 intrusive events.

TWO NAMES, ONE QUANTITY

The meter's own screen and the Nortfr software disagree on labels for exposure. The screen shows **LEA** and **LEC** ; Nortfr exports the same values as **LAE** and **LCE** . They are identical quantities — do not treat a mismatch as an error.

The signal chain

Where the four slots physically happen, and the one path that skips a stage.



The A/C/Z choice and the F/S/I choice are separate, independent stages — which is why every broadband result in this configuration is stored in both an A and a C form. The dashed branch is the whole explanation for peak values: the peak detector taps the weighted signal *before* any averaging, so it can never be lower than a maximum taken after averaging.

The catalogue

Everything stored per run. Symbols are grouped by what they are for, not by their byte order.

Broadband results — one number for the whole run

Each of these exists in an A-weighted and a C-weighted form, giving 22 values. A-weighted values are used for most environmental averages; C-weighted values reveal low-frequency content and are the required form for occupational peak limits.

BROADBAND RESULTS, A AND C		
A / C	WHAT IT IS	WHAT YOU USE IT FOR
AVERAGES		
LAeq LCeq	Energy average over the whole run	The primary result. Feeds nearly every standard.
LA1eq LC1eq	Energy average using the Impulse detector	Comparing against LAeq reveals impulsive character.
MAXIMA — HIGHEST AVERAGED LEVEL		
LAFmax LCFmax	Loudest Fast-averaged moment	Maximum-event criteria, sleep disturbance.

A / C	WHAT IT IS	WHAT YOU USE IT FOR
LASmax LCSmax	Loudest Slow-averaged moment	Usually the lowest of the three maxima. Legacy comparisons.
LAImax LCImax	Loudest Impulse-averaged moment	Impacts, hammering, door slams.
MINIMA		
LAFmin LCFmin	Quietest Fast-averaged moment	Sanity check on background. Prefer LA90 for reporting.
LASmin LCSmin	Quietest Slow-averaged moment	Rarely reported.
LAImin LCImin	Quietest Impulse-averaged moment	Rarely reported.
PEAKS – INSTANTANEOUS, NO AVERAGING		
LAPeak LCpeak	Highest instantaneous pressure	LCpeak carries the hearing-damage limits. Never report peak A-weighted for that purpose.
EXPOSURE		
LAE LCE	All the run's energy as if it lasted 1 second	Comparing events of unequal duration.
LAIE LCIE	Same, Impulse-detected	Specialist use.

Statistical percentiles — the shape of the distribution

A single average hides whether a place was steadily noisy or mostly quiet with occasional lorries. Percentiles recover that. **LA90** means "the level exceeded 90% of the time" — the quiet floor the sound rarely drops below, which is why it is the conventional **indicator of background sound**. It is a well-established proxy for the background, not a direct measurement of it.

PERCENTILES – 8 A-WEIGHTED AND 8 C-WEIGHTED		
SYMBOL	EXCEEDED THIS % OF THE TIME	CONVENTIONAL ROLE
LA0.1 / LA1	0.1% / 1%	Near-peak events, only just below the maximum.
LA5	5%	Loud events.
LA10	10%	Intrusive events. Traditional traffic-noise index.
LA50	50%	Median — the typical level.
LA90	90%	Background sound indicator. The BS 4142 reference value.
LA95 / LA99	95% / 99%	Deep background, approaching the noise floor.

Spectra — 18 tables of 36 frequency bands

Eighteen of the results above also come with a spectrum showing *which frequencies* produced them, in 1/3-octave bands from 6.3 Hz to 20 kHz. Spectral symbols use a lowercase `f` : `Lfeq` is the spectrum behind the equivalent level, `LfFmax` the spectrum behind the Fast maximum. Spectra are what you need to identify a source (a hum at 100 Hz, a fan at 400 Hz) or to design mitigation, since barriers and glazing perform very differently by frequency.

The 18 tables are `Lfeq`, `LfE`, `LfFmax`, `LfFmin`, `LfSmax`, `LfSmin`, `LfIeq`, `LfImax`, `LfImin`, `LfIE`, and the eight `LfF` percentiles. Note that this is fewer than the 38 broadband results — there is no spectrum for every scalar.

SPECTRA ARE UNWEIGHTED

The lowercase `f` identifies a frequency-band spectrum. In these files the stored band levels carry no frequency weighting, and there is no separate A-weighted and C-weighted spectrum — one table serves both. The weighting is applied per band at the moment you sum it, which is why `Lfeq` yields `LAeq` or `LCeq` depending only on which curve you choose. Spectra are also absent from the shorter 1069-byte files, which are too small to contain them (54 of the 527 runs in this archive).

The one-second profile — five channels, every second

PROF CHANNELS		
FIELD	CHANNEL	WHAT IT IS
0	LAFspl	The Fast-weighted level itself, second by second. The statistical descriptors come from this.
1	LAeq,1s	Energy average within each single second.
2	LAFmax,1s	Loudest Fast-averaged moment within each second.
3	LAE,1s	Exposure per second. Numerically equal to <code>LAeq,1s</code> , since the reference period is also one second.
4	LApeak,1s	Instantaneous crest within each second.

This is what draws the time-history chart, and what lets you say *when* something happened rather than only how loud the run was overall.

Where the numbers come from

Three layers: what the meter measured, what it stored, and what this application computes on top.

Layer 1 — the meter's own results, read straight from the file

Everything in the catalogue above is computed inside the instrument and written to the SD card as 16-bit little-endian integers at fixed byte offsets. Decoding is a single formula:


```
dB = uint16_le / 128 - 20 applies to every scalar, spectrum and profile value
```

The scalar results sit at `0x03c1` – `0x03ef`, the percentiles at `0x0408` – `0x0426`, and the 18 spectra from `0x0428` onward in the normal 2653-byte files. These are the **authoritative** values: they come from the instrument's own detectors running at full sample rate, not from anything reconstructed afterwards.

That includes the exposure levels. `LAE` and `LCE` are stored outright, at `0x03d1` and `0x03eb` — they are measured, not calculated from duration.

Layer 2 — deriving broadband levels from a spectrum

You can rebuild a broadband A- or C-weighted level from a spectrum by applying the weighting curve to each band and energy-summing:

```
LAeq = 10 · log10( Σ 10^((Lf_band + A_weight_band) / 10) )
```

This is worth knowing because it is an excellent independent check: on normal 2653-byte files, `Lfeq` and `LfE` reproduce the stored scalars to within a few hundredths of a decibel. Agreement is close but *not* guaranteed across every historical file variant, so treat a mismatch as a prompt to inspect the file rather than proof of a decode bug — and prefer the stored scalar either way. It does not work at all for the max and min spectra, for a reason explained in the traps below.

Layer 3 — what this application computes

Session `LAeq` — two of them

The session list and CSV export show an **equal-weight** energy average of the per-run `LAeq`. Reports compute a **duration-weighted** one, which is the more correct figure when runs differ in length. For runs of equal length — the usual case — they agree exactly. If you see two slightly different session levels for the same date, this is why.

Session `LA10` / `LA90`

Pools every run's one-second samples into one distribution, then takes the percentile of that. The best estimate available from the exported profile, and far better than averaging run percentiles — but an estimate. The per-run values stored by the meter stay authoritative.

Exposure cross-check

`LAE` and `LCE` are read from the file, not computed. But `LAeq ≈ LAeq + 10 · log10(duration)` should hold, which makes it a free validation test — +29.5 dB for a 900-second run.

Time above 85 dB

The percentage of the run spent at or above 85 dB, as an at-a-glance risk indicator. Read it as indicative — see the caveat below.

Traps that catch everyone

Each of these produces a plausible-looking number that is wrong, which is what makes them dangerous.

TRAP 01

Expecting peak and max to be close

They are different physical quantities and the gap is often 10–20 dB. From one real 15-minute run:

LASmax 78.95, LAFmax 80.41, LAImax 84.05, LApeak 92.38 — rising steadily as the averaging window shrinks to nothing. Across ten runs on the same day the peak-minus-max gap ranged from 8.8 to 18.1 dB.

For scale: a steady sine wave has a peak-to-RMS difference of exactly 3.01 dB. That is a useful yardstick, not a floor — a waveform of constant magnitude approaches 0 dB, and real transient signals go far above it. A large gap means impulsive content the averaging has smeared out; it does not correspond to any fixed minimum.

TRAP 02

Averaging percentiles across runs

Percentiles do not compose linearly. The LA90 of two runs combined is *not* the mean of their two LA90 values, especially when the runs differ in length. Averaging the percentiles gives an answer that looks reasonable and is quietly wrong. Pool all the one-second samples and recompute instead: that is the correct method for the sampled series you have, though it still will not exactly reproduce the meter's own full-rate detector.

TRAP 03

Energy-summing a max or min spectrum

Rebuilding a broadband level from LfFmax overestimates LAFmax, and from LfFmin underestimates LAFmin — in one real run by +1.3 dB and –2.5 dB respectively. This is not a decoding error. Each band's maximum occurs at a *different instant*, so summing the per-band peak envelope describes a moment that never existed. Only the time-coherent spectra, Lfeq and LfE, can be summed this way.

TRAP 04

Using a peak value A-weighted for hearing risk

Peak limits in occupational noise law are specified **C-weighted**. The reason is not that impulses are inherently low-frequency — an impulse is broadband — but that A-weighting heavily attenuates low-frequency content, so an A-weighted peak discards pressure that contributes to hearing-damage risk. C-weighting retains most of it.

In one real run LCpeak was 95.53 against LApeak 92.38. In that particular run the impulsive content clearly was low-frequency: C-weighted, the impulse detector sat 11.4 dB above Fast, against only 3.6 dB A-weighted. Reporting LApeak against a LCpeak limit understates the risk.

Trusting a level reconstructed from the chart data

The chart profile is downsampled by taking the maximum of each window, so expanding it back gives a series biased high. It is correct for chart *shape* and misleading as a level. Always prefer the instrument's own scalar results, or the full one-second series, over anything derived from the display profile. The "time above 85 dB" figure is computed this way and should be read as indicative only.

GENERAL RULE

When two sources disagree, the instrument's own stored scalar wins. It was computed by dedicated detectors at full sample rate; everything else is a reconstruction from a summary.

Which standard wants which number

The standards choose their metrics deliberately. Substituting a similar-looking one changes the finding.

METRIC BY ASSESSMENT CONTEXT		
CONTEXT	PRIMARY METRICS	WHY THAT ONE
BS 4142:2014+A1:2019 industrial & commercial sound affecting residents	L _{Aeq} , LA ₉₀	The measured specific sound level is not the number you assess. Character corrections for tonality, impulsivity and intermittency are added first to give a rating level , and that is compared against the background LA ₉₀ — then interpreted in context. Skipping the corrections understates the impact.
Control of Noise at Work 2005 hearing protection	L _{EP,d} · LC _{peak}	Two independent risks: daily personal noise exposure, normalised to 8 hours (80 / 85 / 87 dB) and single-impulse damage (135 / 137 / 140 dB, C-weighted). The exposure figure is a personal dose, not a plain L _{Aeq} of one measurement.
Noise Act 1996 night-time nuisance	L _{Aeq}	Applies to 23:00–07:00, so run start times matter as much as levels. There is no single fixed threshold to compare a 15-minute L _{Aeq} against: the permitted level depends on the underlying sound and is assessed <i>inside</i> the complainant's dwelling.
Statutory nuisance, EPA 1990 s.79	L _{Aeq} , LA ₁₀ , LA ₉₀	No numeric threshold in law. The case is built on level above background, character, timing and duration.
BS 8233:2014 internal levels in dwellings	L _{Aeq,16h} · L _{Aeq,8h}	Separate daytime and night-time internal criteria — 35 dB L _{Aeq,16h} for living rooms, 30 dB L _{Aeq,8h} for bedrooms at night.
Sleep disturbance	LA _{Fmax}	Waking is driven by individual loud events, not by an average.

PRACTICAL CONSEQUENCE

Because BS 4142 turns on a *difference*, the background measurement deserves as much care as the source measurement. Around +5 dB above background indicates an adverse impact and around +10 dB a significant adverse impact, so those two findings sit only 5 dB apart. An error of a few decibels in the background — a run taken at the wrong time of day, or with the wrong sound present — is enough to change the conclusion.

ON THE WEATHER FIGURES

Wind and temperature shown against a session are pulled from a public archive weather service for the site's coordinates, not measured at the microphone. They are useful context — wind especially affects propagation and can invalidate a measurement — but third-party archive weather is **not acceptable as the meteorological evidence** required when submitting sound monitoring or assessments to the environment agencies. Their joint implementation document for BS 4142 — Environment Agency, Natural Resources Wales and the Northern Ireland Environment Agency — states that data from a third-party weather station is not acceptable, and requires wind speed and direction measured at the monitoring location with an anemometer or logging weather station. That requirement governs environmental permitting submissions specifically rather than every BS 4142 assessment, but treat the figures in this app as context only regardless.

Worked example

One real run, 15 minutes, verified against the meter's own screen readings.

2026-08-12, RUN 2 · 21:31:39 · 900 S		
METRIC	VALUE	READING IT
L _{Aeq}	61.05	The energy-average level for the whole 15 minutes.
L _{Ceq}	71.21	10.2 dB above the A value — substantial low-frequency content.
L _{A10}	62.3	Intrusive events sat only 1.3 dB above the average: no dominant events.
L _{A90}	56.1	Background. 4.9 dB below the average.
L _{AFmax}	80.41	Loudest event, 19.4 dB above the average.
L _{Apeak}	92.38	12.0 dB above the maximum — moderately impulsive.
L _{Cpeak}	95.53	The value to compare against occupational peak limits.
L _{AE}	90.59	Read from the file. Cross-check: $61.05 + 10 \cdot \log_{10}(900) = 61.05 + 29.54 = 90.59$. ✓

The narrow L_{A10} – L_{A90} spread of 6.2 dB says this was a fairly steady noise climate, while the 12 dB peak-to-max gap says sharp transients were still present. Steady overall but punctuated by impulses — that is what a bare L_{Aeq} of 61 dB cannot tell you, and what an assessment turns on.

ALWAYS SANITY-CHECK

Three quick tests catch most problems. Does $LAE \approx LA_{eq} + 10 \cdot \log_{10}(\text{duration})$? Do the maxima order $LAS_{max} \leq LAF_{max} \leq LAI_{max} \leq LA_{peak}$? Does $LA_{90} \leq LA_{50} \leq LA_{10}$?

These are plausibility checks, not proofs. In this project's 527-run archive the maxima ordering holds for about 94% of usable runs; of the 27 exceptions, eight sit within 0.5 dB — indistinguishable from detector dynamics and rounding on a near-steady signal, where the three detectors legitimately converge. Most of the larger inversions belong to aborted runs with a truncated profile file. A failed check means *look at this run*, not *the decoding is broken*.

Sources, and how far to trust this

Different claims in this guide rest on very different strengths of evidence. Knowing which is which matters more than a flat list of references.

Tier 1 — measured here, and reproducible

Everything about the binary format, the decode, the value counts, and the run-9 figures was verified directly against the project's own data: the SD-card files, the meter's screen readings, and the Nortfr export. These are the strongest claims in the document because you can re-run them — `python3 test_modules.py` checks them against the reference dataset, including cell-exact parity with both Nortfr workbooks. The corpus statistics — 527 GLOB files, the 94% detector ordering, and the 4.933% of one-second values that *used to* differ from Nortfr by 0.1 dB before that was traced and corrected — come from scanning the archive itself.

Tier 2 — checked against public authoritative guidance

The regulatory figures and method descriptions were checked against free, official material, listed below. This is good evidence, but see the limitations.

SOURCES CONSULTED	
SOURCE	USED FOR
Method implementation document (MID) for BS 4142 Environment Agency, Natural Resources Wales & NIEA	Specific sound level, rating level and the character corrections; background LA_{90} ; the requirement to measure weather at the monitoring position. Its requirements bind environmental permitting submissions, not all BS 4142 work.
Noise at work: guidance for employers HSE	The 80 / 85 / 87 dB(A) exposure action and limit values and the 135 / 137 / 140 dB(C) peak values.
Noise nuisances: how councils deal with complaints GOV.UK	Night hours (23:00–07:00), and that the permitted level depends on the underlying sound and is assessed inside the complainant's dwelling.

SOURCE	USED FOR
Nor140 sound level meter specification Norsonic	Simultaneous A and C (or A and Z) measurement; parallel F/S/I detectors; 1/3-octave analysis; the fixed percentile set; configurable profile intervals; IEC 61672-1 Class 1 conformance.
BS 8233:2014 paywalled — not reviewed	The internal guideline values quoted for dwellings: 35 dB LAeq,16h for living rooms and daytime bedrooms, 30 dB LAeq,8h for bedrooms at night, 40 dB for dining rooms, and 50–55 dB LAeq,T for external amenity space. These are the widely cited Table 4 figures; they were not checked against the purchased standard.
BS 4142:2014+A1:2019 paywalled — not reviewed	The significance indications quoted (around +5 dB above background indicating an adverse impact, around +10 dB a significant adverse impact). The MID above covers the method; these indications come from the standard itself and were not read directly.
IEC 61672-1 paywalled — not reviewed	Sound level meter terminology. Referenced as the instrument's conformance standard; not read in full.

Two details worth noting from that check. The meter's fixed statistical set is seven percentiles plus one user-selectable, which is consistent with the eight stored per weighting here — the 0.1% value being the user-chosen one. And the profile interval is configurable across a wide range, which is why this guide says "configured here for one sample per second" rather than treating one second as inherent.

Tier 3 — interpretation, not classification

Descriptive phrases used to make the numbers readable — "moderately impulsive", "intrusive events", "a fairly steady noise climate" — are explanatory language, not formal categories from any standard. Do not carry them into an assessment as if they were defined terms.

LIMITATIONS

The full BS and IEC standards are paywalled, so this was **not** a licensed clause-by-clause compliance audit. It is a field reference for understanding what the instrument records and how the numbers relate to each other.

For a formal assessment, work from the applicable standard itself. Where this guide and the standard disagree, the standard is right.

Values throughout are from the project's own reference dataset for 2026-08-12, cross-checked against the meter screen and the Norsonic Nortfr export. As of 2026-08-17 the generated GLOBAL and PROFILE workbooks match that Nortfr export in *every cell* — all 38 scalars, 648 spectral values and 4,500 one-second samples — so a difference against a vendor export is now a signal worth investigating rather than an expected artifact. Binary offsets, decode derivation and the open questions about unusual file variants are documented in `NOR140_handoff.md`.